One relation behind the calculators

$$\sigma_{ab}(i) \approx \sum_s I_s\, D_{ab}(\mathbf{r}_{is}), \qquad D_{ab} = \partial_a \partial_b \frac{1}{\rho} = \frac{3\hat{\rho}_a\hat{\rho}_b - \delta_{ab}}{\rho^3}$$

each contributor is a named, equation-bearing projection of it — the question is whether ours correspond.

where $\sigma_{ab}(i)$ — the shielding tensor at nucleus i (what DFT gives, what we predict); I_s — how strongly source s contributes (its calibration coefficient, geometry to ppm); D_{ab} — the dipolar kernel for the nucleus-to-source step $\mathbf{r}_{is}$; ρ, $\hat{\rho}$ — that step's length and direction; $(3\cos^2\theta - 1)/\rho^3$ — strongest along the axis, zero at the magic angle ($\cos^2\theta = \frac{1}{3}$).

classes: process · question · data source · evaluative tool · model · finding

STAGE 1 · static — mutants

asks distribution in space (dihedral patterns?); do the contributors correspond to a law? (ridge fit; reflect physics)

data 720 WT+ALA pairs; DFT $\Delta\sigma$

tools extractor; ridge; contribution; stats

finds static contributors, per element / atom-type

carry forward →

STAGE 2 · in-frame — 1P9J

asks can they operate as a model? do they reflect the physics in-frame? (do they predict shift?)

data 1P9J trajectory; per-frame DFT; + trajectory-only metrics

tools extractor; contribution; stats; the D_{ab}-sum

finds in-frame laws — the D_{ab}-sum recovers the tensor

builds contributors-as-model (the D_{ab}-sum)

freeze + transfer →

STAGE 3 · the model

asks does the model help predict shift? favour secondary structure / residues / regions (IUPAC)? what geometry drives it?

data BMRB measured shifts

tools ablation (quality / pLDDT; Model 1 in/out; calculators in/out)

builds Model 1 (e3nn, on DFT) → Model 2 (big, vs shifts)

finds predicted shift + what the model favours

Model 1 trains on the Stage 1 + Stage 2 DFT (720 statics + 1P9J frames), then is frozen and carried into Stage 3 — the clean physics never sees the noisy experiment.